

- This exam contains 8 pages (including the front and back of this cover page) and 9 problems. Check to see if any pages are missing.
- This is the final exam and will count for 40% of your final grade. You have 120 minutes (1:15 PM - 3:15 PM) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 440 points total in this exam, as well as 2 bonus questions worth a total of 20 possible points.

Page:	3	4	5	6	7	Total
Points:	40	140	120	60	80	440
Score:						

Bonus:	
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Review Day Question Bonus:	
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1.

For readability in the key, correct answers are colored in red, e.g. **Correct Answer**, while incorrect answers are marked with a strike, e.g. ~~Incorrect Answer~~.

- (a) 4 points If λ is a scalar such that $Ax = \lambda x$ for some $x \in \mathbb{R}^n$, λ is an eigenvalue of A .

True~~False~~

- (b) 4 points $\det(AB) = \det(A) - \det(B)$.

~~True~~**False**

- (c) 4 points If $A = P \begin{bmatrix} 3 & 0 \\ 0 & 8 \end{bmatrix} P^{-1}$, then $A^{2019} = P \begin{bmatrix} 3^{2019} & 0 \\ 0 & 8^{2019} \end{bmatrix} P^{-1}$.

True~~False~~

- (d) 4 points If A is diagonalizable, then A is invertible.

~~True~~**False**

- (e) 4 points If a vector x is orthogonal to every vector in a subspace W , then $x \in W^\perp$.

True~~False~~

- (f) 4 points If A is any $n \times m$ matrix, then $(\text{Col}(A))^\perp = \text{Nul}(A)$.

~~True~~**False**

- (g) 4 points If $\{v_1, \dots, v_p\}$ is a linearly independent set in $\mathbb{R}^n$, then it is also an orthogonal set.

~~True~~**False**

- (h) 4 points If $\{v_1, \dots, v_p\}$ is an orthogonal basis for a subspace W of $\mathbb{R}^n$, then for any $y \in W$, we have $y = \frac{y \cdot v_1}{v_1 \cdot v_1} v_1 + \dots + \frac{y \cdot v_p}{v_p \cdot v_p} v_p$.

True~~False~~

- (i) 4 points The best approximation to y by elements of a subspace W is given by the vector $y - \text{proj}_W(y)$.

~~True~~**False**

- (j) 4 points If $x \in \mathbb{R}^n$ is a nonzero vector and W is a nonzero subspace of $\mathbb{R}^n$, then $\text{proj}_W(x) \neq 0$.

~~True~~**False**

2. 60 points Show that if A^2 is the zero matrix, then the only eigenvalue of A is 0.

Suppose that $A^2 = 0$ for some $n \times n$ matrix A and let λ be an arbitrary eigenvalue of A . Since we have supposed that λ is an eigenvalue of A , then by definition it must be true that $Av = \lambda v$ for some nonzero vector v . So, we have

$$\begin{aligned} 0 &= A^2v = AA v \\ &= A(\lambda v) \\ &= \lambda Av = \lambda^2 v. \end{aligned}$$

But, since v is not the zero vector, the statement $0 = \lambda^2 v$ then tells us that $\lambda = 0$ must hold. So, since λ was taken as an arbitrary eigenvalue of A , our conclusion that $\lambda = 0$ holds irrespective of our choice of eigenvalue λ . Therefore, it must be the case that $\lambda = 0$ is the only eigenvalue of A .

3. 80 points Find the characteristic polynomial and the eigenvalues of the matrix $\begin{bmatrix} -3 & 4 \\ 6 & 2 \end{bmatrix}$.

To find the characteristic polynomial of the matrix $\begin{bmatrix} -3 & 4 \\ 6 & 2 \end{bmatrix}$, recall that the characteristic polynomial of a matrix A is simply the polynomial $\det(A - \lambda I)$. So, we first compute:

$$\begin{bmatrix} -3 & 4 \\ 6 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} (-3 - \lambda) & 4 \\ 6 & (2 - \lambda) \end{bmatrix}.$$

Now, we compute the determinate of the matrix above:

$$\begin{aligned} \det \left(\begin{bmatrix} (-3 - \lambda) & 4 \\ 6 & (2 - \lambda) \end{bmatrix} \right) &= (-3 - \lambda)(2 - \lambda) - (4)(6) \\ &= -6 + \lambda + \lambda^2 - 24 \\ &= \lambda^2 + \lambda - 30. \end{aligned}$$

So, the characteristic polynomial of the matrix $\begin{bmatrix} -3 & 4 \\ 6 & 2 \end{bmatrix}$ is $\lambda^2 + \lambda - 30$. To find the eigenvalues of this matrix, we need only solve $\lambda^2 + \lambda - 30 = 0$. Factoring, we have

$$\lambda^2 + \lambda - 30 = (\lambda - 5)(\lambda + 6).$$

So, the eigenvalues of the matrix $\begin{bmatrix} -3 & 4 \\ 6 & 2 \end{bmatrix}$ are $\lambda_1 = 5$ and $\lambda_2 = -6$.

4. 60 points Suppose A has n linearly independent eigenvectors. Prove that A^T also has n linearly independent eigenvectors. (*HINT: Use the Diagonalization Theorem*)

Suppose that A has n linearly independent eigenvectors, and denote them by $v_1, \dots, v_n$. Then, by the Diagonalization Theorem, we have

$$A = PDP^{-1} = \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix} D \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix}^{-1}$$

where D is a diagonal matrix whose i^{th} entry on the diagonal is the eigenvalue of A corresponding to the eigenvector v_i . So, since $D^T = D$ for any diagonal matrix D and $(M^{-1})^T = (M^T)^{-1}$ for any invertible matrix M , we have

$$\begin{aligned} A^T &= (PDP^{-1})^T \\ &= (P^{-1})^T D^T P^T \\ &= (P^T)^{-1} D P^T. \end{aligned}$$

Writing $R = (P^T)^{-1}$, this then becomes $A^T = RDR^{-1}$. Moreover, as R is an invertible matrix (as the transpose of an invertible matrix is always invertible), the Diagonalization Theorem yields that the columns of R must be linearly independent eigenvectors of A . Therefore, since R has n columns, we may conclude that A^T has n linearly independent eigenvectors as well.

5. 60 points Let $T : \mathbb{P}_2 \rightarrow \mathbb{P}_1$ be the transformation that maps a polynomial, $p(t)$, to its second derivative, $p''(t)$. Find the matrix for T relative to the basis $\mathcal{B} = \{1, t, t^2\}$ for $\mathbb{P}_2$ and the basis $\mathcal{D} = \{1, t\}$ for $\mathbb{P}_1$.

As $T : \mathbb{P}_2 \rightarrow \mathbb{P}_1$ takes a polynomial $p(t) \in \mathbb{P}_2$ to its second derivative $p''(t) \in \mathbb{P}_1$, let's see what T does to a generic polynomial $p(t) = a + bt + ct^2 \in \mathbb{P}_2$ where $a, b, c \in \mathbb{R}$ are arbitrary constants. That is, let's observe that $T(p(t)) = \frac{d^2}{dt^2}(a + bt + ct^2) = 2c$. So, using the basis $\mathcal{B} = \{1, t, t^2\}$ for $\mathbb{P}_2$ and the basis $\mathcal{D} = \{1, t\}$ for $\mathbb{P}_1$, we have that the matrix for T relative to the bases $\mathcal{B}$ and $\mathcal{D}$ is the matrix

$$M \text{ such that if } [p(t)]_{\mathcal{B}} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \text{ then } M[p(t)]_{\mathcal{B}} = [T(p(t))]_{\mathcal{D}} = [2c]_{\mathcal{D}} = \begin{bmatrix} 2c \\ 0 \end{bmatrix}.$$

Using the formula for M provided in Section 5.4 of the textbook, we have

$$\begin{aligned} M &= \begin{bmatrix} [T(b_1)]_{\mathcal{D}} & [T(b_2)]_{\mathcal{D}} & [T(b_3)]_{\mathcal{D}} \end{bmatrix} = \begin{bmatrix} [T(1)]_{\mathcal{D}} & [T(t)]_{\mathcal{D}} & [T(t^2)]_{\mathcal{D}} \end{bmatrix} \\ &= \begin{bmatrix} [0]_{\mathcal{D}} & [0]_{\mathcal{D}} & [2]_{\mathcal{D}} \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$

NOTE: Since I didn't specify the order of the basis elements in $\mathcal{B}$ and $\mathcal{D}$, I gave full points for a correct answer produced using any valid choice of ordering of these basis elements.

6. 60 points Let $v_1 = \begin{bmatrix} 2 \\ -1 \\ -3 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix}$, and $z = \begin{bmatrix} 3 \\ -7 \\ 2 \\ 3 \end{bmatrix}$. Let $W = \text{span}\{v_1, v_2\}$ and find the best approximation to z by elements of W .

Recall that the Best Approximation Theorem (Theorem 9 of Section 6.3 in your textbook) states that for any subspace W of $\mathbb{R}^n$ and any vector $y \in \mathbb{R}^n$, the vector $\hat{y} = \text{proj}_W(y)$ is the closest point in W to y in the sense that $\|y - \hat{y}\| < \|y - v\|$ for all vectors $v \in W$ satisfying $v \neq \hat{y}$.

As such, we know that the best approximation to z by elements of $W = \text{span}\{v_1, v_2\}$ is given by $\text{proj}_W(z)$. To compute $\text{proj}_W(z)$, we must first confirm that $\{v_1, v_2\}$ is an orthogonal set. To do so, we simply compute

$$v_1 \cdot v_2 = (2)(1) + (-1)(1) + (-3)(0) + (1)(-1) = 2 - 1 + 0 - 1 = 0.$$

So, $\{v_1, v_2\}$ is indeed an orthogonal set. Moreover, as $W = \text{span}\{v_1, v_2\}$, the set $\{v_1, v_2\}$ is also an orthogonal basis for W . Hence, we have $\text{proj}_W(z) = \frac{z \cdot v_1}{v_1 \cdot v_1} v_1 + \frac{z \cdot v_2}{v_2 \cdot v_2} v_2$, and the best approximation to z by elements of W is given by

$$\begin{aligned} \text{proj}_W(z) &= \frac{z \cdot v_1}{v_1 \cdot v_1} v_1 + \frac{z \cdot v_2}{v_2 \cdot v_2} v_2 \\ &= \frac{(3)(2) + (-7)(-1) + (2)(-3) + (3)(1)}{(2)^2 + (-1)^2 + (-3)^2 + (1)^2} v_1 + \frac{(3)(1) + (-7)(1) + (2)(0) + (3)(-1)}{(1)^2 + (1)^2 + (0)^2 + (-1)^2} v_2 \\ &= \frac{2}{3} \begin{bmatrix} 2 \\ -1 \\ -3 \\ 1 \end{bmatrix} + \frac{-7}{3} \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ -3 \\ -2 \\ 3 \end{bmatrix}. \end{aligned}$$

7. 80 points Let $v_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 8 \\ 5 \\ 6 \end{bmatrix}$, and write $W = \text{span}\{v_1, v_2\}$. Use the Gram-Schmidt process to find an orthogonal basis for W .

Recall that the Gram-Schmidt Process (Theorem 11 of Section 6.4 in your textbook) states that if $\{x_1, \dots, x_p\}$ is a basis for a nonzero subspace W of $\mathbb{R}^n$, then the set $\{u_1, \dots, u_p\}$ is an orthogonal basis for W , where

$$\begin{aligned} u_1 &= x_1 \\ u_2 &= x_2 - \frac{x_2 \cdot u_1}{u_1 \cdot u_1} u_1 \\ u_3 &= x_3 - \frac{x_3 \cdot u_1}{u_1 \cdot u_1} u_1 - \frac{x_3 \cdot u_2}{u_2 \cdot u_2} u_2 \\ &\vdots \\ u_p &= x_p - \frac{x_p \cdot u_1}{u_1 \cdot u_1} u_1 - \frac{x_p \cdot u_2}{u_2 \cdot u_2} u_2 - \dots - \frac{x_p \cdot u_{p-1}}{u_{p-1} \cdot u_{p-1}} u_{p-1}. \end{aligned}$$

So, with this in mind, note that the set $\{v_1, v_2\}$ given in the statement of the problem is a basis for the subspace $W = \text{span}\{v_1, v_2\}$, as $\{v_1, v_2\}$ is a linearly independent set. So, since W is a nonzero subspace of $\mathbb{R}^3$ and $\{v_1, v_2\}$ is a basis for W , we may use the Gram-Schmidt process on $\{v_1, v_2\}$ to produce an orthogonal basis $\{u_1, u_2\}$ of W by setting:

$$u_1 = v_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} \quad \text{and} \quad u_2 = v_2 - \frac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1 = v_2 - \frac{9}{5} v_1 = \begin{bmatrix} 13/5 \\ 5 \\ 39/5 \end{bmatrix} = \begin{bmatrix} 2.6 \\ 5 \\ 7.8 \end{bmatrix}.$$

8. 10 points (bonus) Let W be a subspace of $\mathbb{R}^n$. Show that if $x \in \mathbb{R}^n$ is a vector which satisfies both $x \in W$ and $x \in W^\perp$, then $x = 0$.

Suppose W is a subspace of $\mathbb{R}^n$ and $x \in \mathbb{R}^n$ is a vector which satisfies both $x \in W$ and $x \in W^\perp$. Recall that, by definition, $W^\perp$ is the subspace of $\mathbb{R}^n$ defined by

$$W^\perp = \{v \in \mathbb{R}^n : v \cdot w = 0 \text{ for all } w \in W\}.$$

So, if $x \in W^\perp$, then $x \cdot w = 0$ for all $w \in W$. But, since we have also supposed that $x \in W$ as well, this yields that x must satisfy

$$x \cdot x = 0.$$

By Theorem 1 of Section 6.1 in your textbook, a vector $u \in \mathbb{R}^n$ satisfies $u \cdot u = 0$ if and only if $u = 0$. Therefore, we may conclude that if W is a subspace of $\mathbb{R}^n$ and $x \in \mathbb{R}^n$ satisfies both $x \in W$ and $x \in W^\perp$, then $x = 0$.

9. 10 points (bonus) Let A be an $m \times n$ matrix such that $A^T A$ is invertible. Show that the columns of A are linearly independent.

(WARNING: Since A may not be a square matrix, you may **not** assume that A is invertible!)

Let A be an $m \times n$ matrix such that $A^T A$ is invertible. We want to show that the columns of A are linearly independent.

To do so, let us note that since $A^T A$ is invertible, the homogeneous system $A^T A x = 0$ has only the trivial solution $x = 0$. Consequentially, since any vector $x \in \text{Nul}(A)$ satisfies $Ax = 0$, it must be the case that $\text{Nul}(A) = \{0\}$, since the inclusion of any nonzero vector y in $\text{Nul}(A)$ would then produce a nontrivial solution to $A^T A x = 0$. That is, if there existed a vector $y \neq 0$ belonging to $\text{Nul}(A)$, we would have $A^T A y = A^T(0) = 0$ for some nonzero vector y , contradicting the invertibility of $A^T A$.

So, writing $A = [a_1 \ \dots \ a_n]$, where $a_1, \dots, a_n \in \mathbb{R}^m$ are the columns of A , let us recall that

$$Ax = [a_1 \ \dots \ a_n] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 a_1 + \dots + x_n a_n.$$

Since $Ax = 0$ has only the trivial solution $x = 0$, this then says that $x_1 a_1 + \dots + x_n a_n = 0$ if and only if $x_1 = \dots = x_n = 0$. Since this is precisely the definition of linear independence, we may conclude that the columns of A must be linearly independent whenever $A^T A$ is invertible.